

ELECTIVE-3

MASTER OF COMPUTER APPLICATION (MCA)

BMC035: Software Quality Engineering		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to:		
CO 1	Understand basic concepts of Software Quality along with its documents and process	K2
CO 2	Apply knowledge of Software Quality in various types of software	K3
CO 3	Compare the various reliability models for different scenarios	K4
CO 4	Illustrate the software Quality Planning and Assurance	K2
CO 5	Make use of various testing techniques in software implementation	K3
DETAILED SYLLABUS		3-1-0
Unit	Topic	Proposed Lecture
I	Software Quality: Definition, Software Quality Attributes and Specification, Cost of Quality, Defects, Faults, Failures, Defect Rate and Reliability, Defect Prevention, Reduction, and Containment, Overview of Different Types of Software Review, Introduction to Measurement and Inspection Process, Documents and Metrics.	08
II	Software Quality Metrics Product Quality Metrics: Defect Density, Customer Problems Metric, Customer Satisfaction Metrics, Function Points, In-Process Quality Metrics: Defect Arrival Pattern, Phase-Based Defect Removal Pattern, Defect Removal Effectiveness, Metrics for Software Maintenance: Backlog Management Index, Fix Response Time, Fix Quality, Software Quality Indicators.	08
III	Software Quality Management and Models: Modeling Process, Software Reliability Models: The Rayleigh Model, Exponential Distribution and Software Reliability Growth Models, Software Reliability Allocation Models, Criteria for Model Evaluation, Software Quality Assessment Models: Hierarchical Model of Software Quality Assessment.	08
IV	Software Quality Assurance: Quality Planning and Control, Quality Improvement Process, Evolution of Software Quality Assurance (SQA), Major SQA Activities, Major SQA Issues, Zero Defect Software, SQA Techniques, Statistical Quality Assurance, Total Quality Management, Quality Standards and Processes.	08
V	Software Verification, Validation & Testing: Verification and Validation, Evolutionary Nature of Verification and Validation, Impracticality of Testing all Data and Paths, Proof of Correctness, Software Testing, Functional, Structural and Error-Oriented Analysis & Testing, Static and Dynamic Testing Tools, Characteristics of Modern Testing Tools.	08
Text books: 1. Jeff Tian, Software Quality Engineering (SQE), Wiley-Interscience, 2005; ISBN 0-471-71345 -7 2. Metrics and Models in Software Quality Engineering, Stephen H. Kan, AddisonWesley (2002), ISBN: 0201729156 3. Norman E. Fenton and Shari Lawrence Pfleeger, "Software Metrics" Thomson, 2003 4. Mordechai Ben – Menachem and Garry S.Marliss, "Software Quality", Thomson Asia Pte Ltd, 2003.		

ELECTIVE-4

MASTER OF COMPUTER APPLICATION (MCA)

BMC045: Distributed Database Systems		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course , the student will be able to:		
CO 1	Understand theoretical and practical aspects of distributed database systems.	K2
CO 2	Study and identify various issues related to the development of distributed database system	K3
CO 3	Understand the design aspects of object-oriented database system and related development	K4
CO 4	Equip students with principles and knowledge of distributed reliability.	K3
CO 5	Equip students with principles and knowledge of parallel and object-oriented databases.	K5
DETAILED SYLLABUS		4-0-0
Unit	Topic	Proposed Lecture
I	Introduction: Distributed Data Processing, Distributed Database System, Promises of DDBSS, Problem areas. Distributed DBMS Architecture: Architectural Models for Distributed DBMS, DDMBS Architecture. Distributed Database Design: Alternative Design Strategies, Distribution Design issues, Fragmentation, Allocation.	08
II	Query processing and decomposition: Query processing objectives, characterization of query processors, layers of query processing, query decomposition, localization of distributed data. Distributed query Optimization: Query optimization, centralized query optimization, distributed query optimization algorithms.	08
III	Transaction Management: Definition, properties of transaction, types of transactions, distributed concurrency control: Serializability, concurrency control mechanisms & algorithms, time - stamped & optimistic concurrency control Algorithms, deadlock Management.	08
IV	Distributed DBMS Reliability: Reliability concepts and measures, fault- tolerance in distributed systems, failures in Distributed DBMS, local & distributed reliability protocols, site failures and network partitioning. Parallel Database Systems: Parallel database system architectures, parallel data placement, parallel query processing, load balancing, database clusters.	08
V	Distributed object Database Management Systems: Fundamental object concepts and models, object distributed design, architectural issues, object management, distributed object storage, object query Processing. Object Oriented Data Model: Inheritance, object identity, persistent programming languages, persistence of objects, comparison OODBMS and ORDBMS	08
Text books: M. Tamer OZSU and Patuck Valduriez: Principles of Distributed Database Systems, Pearson Edn. Asia, 2001. 2. Stefano Ceri and Giuseppe Pelagatti: Distributed Databases, McGraw Hill. REFERENCE BOOKS: 1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom: "Database Systems: The Complete Book", Second Edition, Pearson International Edition		

ELECTIVE-5

MASTER OF COMPUTER APPLICATION (MCA)

BMC052: Computer Graphics and Animation		
Course Outcome (CO)		Bloom's Knowledge Level (KL)
At the end of course, the student will be able to understand		
CO 1	Understand the graphics hardware used in field of computer graphics.	K ₂
CO 2	Understand the concept of graphics primitives such as lines and circle based on different algorithms.	K ₂ , K ₄
CO 3	Apply the 2D graphics transformations, composite transformation and Clipping concepts.	K ₄
CO 4	Apply the concepts and techniques used in 3D computer graphics, including viewing transformations, projections, curve and hidden surfaces.	K ₂ , K ₃
CO 5	Perform the concept of multimedia and animation in real life.	K ₂ , K ₃
DETAILED SYLLABUS		3-0-0
Unit	Topic	Proposed Lecture
I	Introduction and Line Generation: Types of computer graphics, Graphic Displays- Random scan displays, Raster scan displays, Frame buffer and video controller, Points and lines, Line drawing algorithms, Circle generating algorithms, Mid-point circle generating algorithm, and parallel version of these algorithms.	08
II	Transformations: Basic transformation, Matrix representations and homogenous coordinates, Composite transformations, Reflections and shearing. Windowing and Clipping: Viewing pipeline, Viewing transformations, 2-D Clipping algorithms- Line clipping algorithms such as Cohen Sutherland line clipping algorithm, Liang Barsky algorithm, Line clipping against non rectangular clip windows; Polygon clipping – Sutherland Hodgeman polygon clipping, Weiler and Atherton polygon clipping, Curve clipping, Text clipping.	08
III	Three Dimensional: 3-D Geometric Primitives, 3-D Object representation, 3-D Transformation, 3-D viewing, projections, 3-D Clipping. Curves and Surfaces: Quadric surfaces, Spheres, Ellipsoid, Blobby objects, Introductory concepts of Spline, Bspline and Bezier curves and surfaces.	08
IV	Hidden Lines and Surfaces: Back Face Detection algorithm, Depth buffer method, A- buffer method, Scan line method, basic illumination models– Ambient light, Diffuse reflection, Specular reflection and Phong model, Combined approach, Warn model, Intensity Attenuation, Color consideration, Transparency and Shadows.	08
V	Multimedia Systems: Design Fundamentals, Back ground of Art, Color theory overview, Sketching & illustration, Storyboarding, different tools for animation. Animation: Principles of Animations, Elements of animation and their use, Power of Motion, Animation Techniques, Animation File Format, Making animation for Rolling Ball, making animation for a Bouncing Ball, Animation for the web, GIF, Plugins and Players, Animation tools for World Wide Web.	08
Suggested Readings: - Hearn D. and Baker M. P., “Computer Graphics C Version”, Pearson Education - Foley, Vandam, Feiner, Hughes, “Computer Graphics principle”, Pearson Education. - Rogers, “Procedural Elements of Computer Graphics”, McGraw Hill - Newman W. M., Sproull R. F., “Principles of Interactive computer Graphics”, McGraw Hill. - Sinha A. N. and Udai A. D., “Computer Graphics”, McGraw Hill. - Mukherjee, “Fundamentals of Computer graphics & Multimedia”, PHI Learning Private Limited. - Vaughan T., “Multimedia, Making IT Work”, Tata McGraw Hill.		